

QSKILL INTERNSHIP

AI & MACHINE LEARNING

SPAM MAIL DETECTOR

Machine Learning Project Report

Project Details	Information
Task	Task 2 – Spam Mail Detector
Dataset	SMS Spam Collection
Samples	5,572
Model	Logistic Regression
Text Representation	TF-IDF
Test Accuracy	97.31%

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1. Objective

The objective of this project is to develop a Machine Learning model that can classify text messages into two categories: Spam and Ham (normal message). The project demonstrates text preprocessing, feature extraction, model training, prediction, and evaluation.

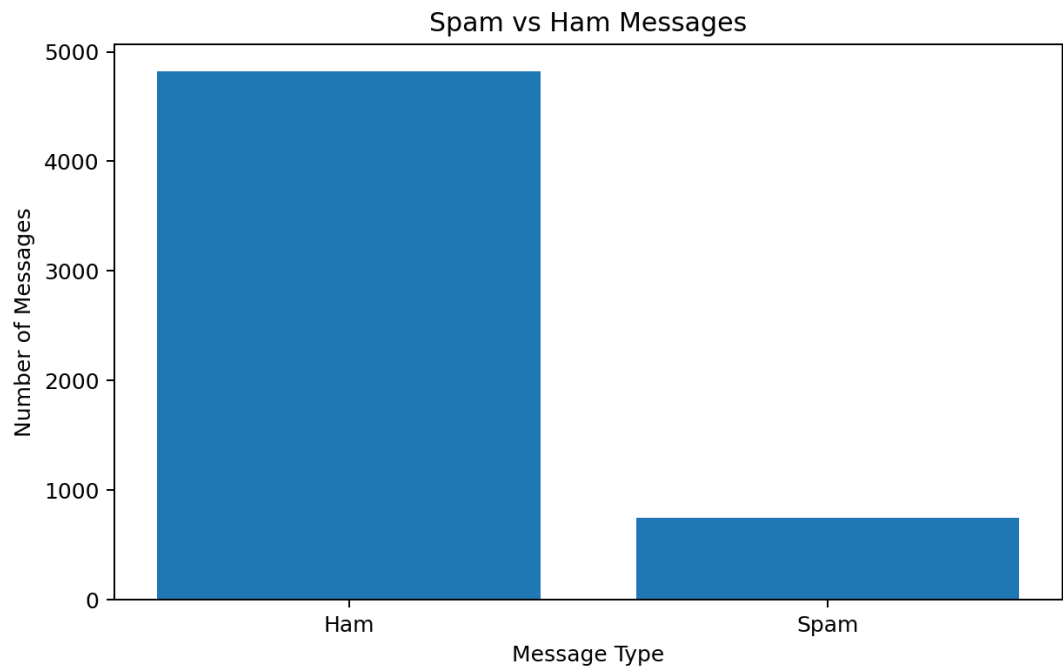
2. Introduction

Spam messages are unwanted messages that may contain advertisements, promotional content, or suspicious offers. A Spam Mail Detector can automatically identify such messages and separate them from normal messages. In this project, text messages are converted into numerical features using TF-IDF and classified using Logistic Regression.

3. Dataset Description

The notebook uses an SMS spam dataset containing 5,572 messages with two columns: label and message. The label identifies whether a message is Ham or Spam.

Dataset Information	Value
Total messages	5,572
Columns	2
Labels	Ham, Spam
Ham messages	4,825
Spam messages	747
Missing values	0



4. Tools and Technologies

Tool / Technology	Purpose
Python	Programming language
Google Colab	Development environment
Pandas	Data loading and analysis

NumPy	Numerical operations
Matplotlib / Seaborn	Visualization
Scikit-learn	Machine Learning
TF-IDF Vectorizer	Converts text into numerical features
Logistic Regression	Classification model

5. Methodology

1. Import required Python libraries.
2. Download and load the SMS dataset.
3. Inspect dataset shape, columns, class counts, and missing values.
4. Separate message text (input) and message labels (target).
5. Split the dataset into training and testing sets using an 80:20 ratio.
6. Convert text into numerical features using TF-IDF.
7. Train a Logistic Regression classifier.
8. Generate predictions for the test messages.
9. Evaluate the model using accuracy, classification report, and confusion matrix.
10. Test the trained model with new messages.

6. Train/Test Split

The dataset was divided into 80% training data and 20% testing data. The notebook recorded 4,457 training messages and 1,115 testing messages. The split used `random_state=42` and stratification by label.

7. Text Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) was used to convert the text messages into numerical features. The resulting training matrix had 4,457 samples and 7,668 features, while the testing matrix had 1,115 samples and 7,668 features.

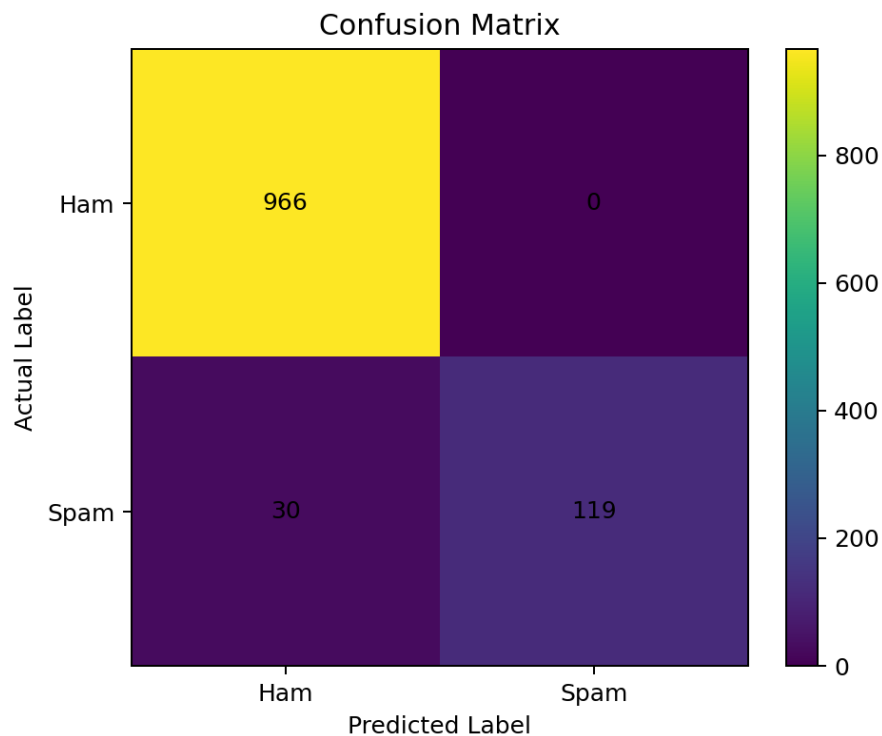
8. Machine Learning Model

Logistic Regression was selected as the classification model. The model was trained using the TF-IDF features with `max_iter=1000`.

9. Model Evaluation

The model achieved an accuracy of 97.31% on the test dataset.

Metric	Ham	Spam
Precision	0.97	1.00
Recall	1.00	0.80
F1-score	0.98	0.89
Support	966	149



The confusion matrix corresponding to the notebook's reported test results is $[[966, 0], [30, 119]]$. This indicates that 966 Ham messages and 119 Spam messages were correctly classified, while 30 Spam messages were classified as Ham.

10. New Message Prediction

Test Message	Prediction
Congratulations! You have won a free prize. Click now!	Spam
Hey, are we meeting tomorrow for the project?	Ham

11. Results

- The dataset contained 5,572 messages with no missing values.
- The model was trained on 4,457 messages and tested on 1,115 messages.
- TF-IDF successfully converted the text messages into numerical features.
- Logistic Regression achieved 97.31% test accuracy.
- The model correctly identified the provided new Spam and Ham examples.

12. Conclusion

The Spam Mail Detector successfully classifies messages as Spam or Ham using Machine Learning. TF-IDF was used to convert text messages into numerical features, and Logistic Regression was used for classification. The model was evaluated using accuracy, classification report, and confusion matrix. The trained model can also predict whether a new message is Spam or Ham.

13. Project File

The implementation is available in the Google Colab notebook: [Spam_Mail_Detector.ipynb](#).